

Strategy Blueprint



Liquidity-Weighted Trend Strategy



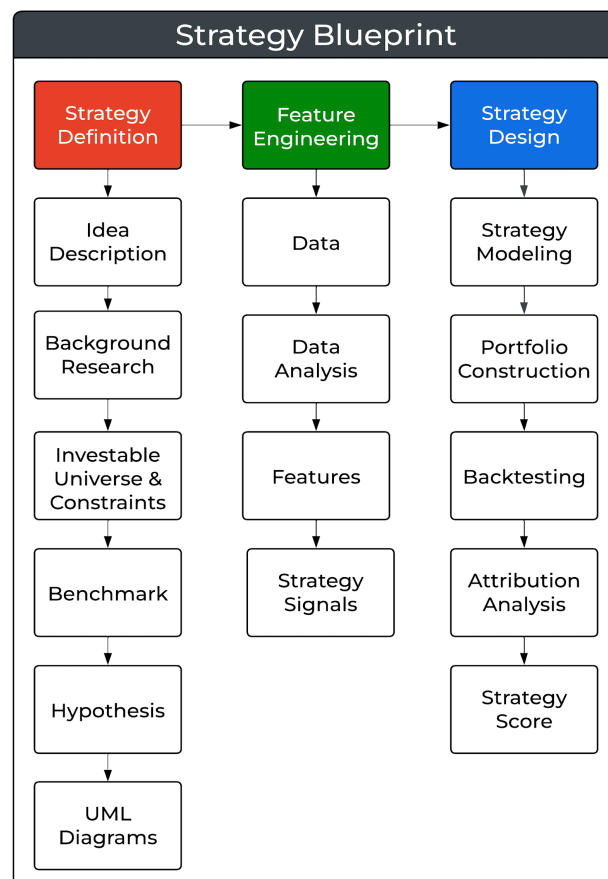
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The Process Flow:

Strategy Definition - Moving from intuition to a **clear, testable hypothesis** with an explicit source of returns.

Feature Engineering - Transforming raw data into **meaningful, hypothesis-driven signals**.

Strategy Design - Translating signals into **portfolio decisions** — selection, sizing, and timing — and validating results through **robust testing and analysis**.



Strategy Blueprint



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Strategy Definition

Idea Description

This strategy explores the persistence of price trends in US equities using a simple but robust signal: the crossover of the 50-day and 200-day Simple Moving Averages (SMA).

The central idea is that stocks experiencing sustained bullish trends tend to continue outperforming the market due to:

- Gradual information diffusion across market participants
- Institutional trading flows that reinforce existing momentum
- Behavioral underreaction by retail investors to new information

Trend persistence is identified through an SMA crossover signal. Stocks are considered bullish when $SMA(50) > SMA(200)$.

Eligible securities are ranked by 3-month traded value, and the 35 or 20 most liquid stocks are selected. If fewer than X stocks satisfy the trend condition, the remaining capital is allocated to the SPY ETF (SPDR S&P 500 ETF Trust) to keep the portfolio fully invested.

Background Research

Momentum and trend-following strategies have been extensively documented in academic financial research.

Returns to Buying Winners and Selling Losers

Date: 01-01-1993

Authors: Narasimhan Jegadeesh and Sheridan Titman

Link: <https://doi.org/10.1111/j.1540-6261.1993.tb04702.x>

- Stocks with strong past performance tend to continue outperforming over short and medium-term horizons.



- One of the most widely replicated anomalies in finance, supporting the foundation of momentum-based strategies.

Time Series Momentum

Date: 01-01-2012

Authors: Tobias Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen

Link: <https://doi.org/10.1016/j.jfineco.2011.11.003>

- Trend-following strategies generate positive returns across equities, commodities, currencies, and bonds.
- Momentum effects persist across global markets, reinforcing the validity of the SMA crossover as a trend-capture mechanism.

Investor Psychology and Security Market Under- and Overreactions

Date: 01-12-1998

Authors: Kent Daniel, David Hirshleifer, and Avanidhar Subrahmanyam

Link: <https://doi.org/10.1111/0022-1082.00077>

- Investor underreaction to new information can lead to persistent price trends, providing a behavioral foundation for the SMA signal.

Trading Costs of Asset Pricing Anomalies

Date: 01-01-2014

Authors: Andrea Frazzini, Ronen Israel, and Tobias J. Moskowitz

Link: <https://doi.org/10.2139/ssrn.2294498>

- Liquidity constraints and transaction costs are critical when implementing anomaly-based strategies.



- This motivates the use of a 3-month traded value filter to ensure the portfolio targets the most accessible and scalable securities.

Investable Universe and Constraints

Universe

- KN US Equity Benchmark Historical Holdings
- US-listed equities only

Strategy Constraints

- Minimum trading history required to compute the 200-day SMA (at least 200 trading days)
- Liquidity filter: securities ranked by 3-month average daily traded value
- Long-only portfolio implementation
- Maximum position size of 20% of the portfolio in any single name
- Top 35 holdings target; remaining capital allocated to SPY when eligible stocks fall below 35

Benchmark

- **KN US Equity Benchmark** — provides the investable universe, daily historical weights, and transparent construction rules
- **SPDR S&P 500 ETF Trust (SPY)**
 - FMP Ticker: SPY
 - ISIN: US78462F1030
 - IPO: 01/29/1993
 - Used as the fallback allocation when fewer than 35 stocks satisfy the trend condition

Comparison against the benchmark enables evaluation of excess returns, Tracking Error, and Active performance / Information Ratio.



Hypothesis

Market Inefficiency

- Price momentum exists due to investor underreaction to new information, causing prices to gradually adjust rather than instantaneously.
- Institutional portfolio adjustments occur gradually, amplifying and extending existing trends.

Behavioral

- Investors systematically underreact to new information, creating persistent price trends.
- Retail investors chase past performance, creating predictable momentum patterns

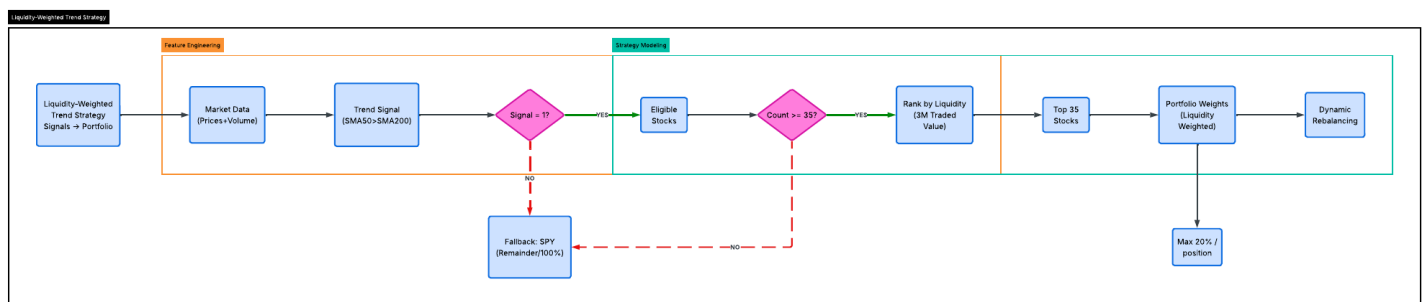
Structural

- Liquidity concentration reinforces momentum among the most actively traded securities.
- Institutional rebalancing flows create temporary but exploitable price pressure.

Null Hypothesis

- The strategy does not outperform the KN US Equity Benchmark on a risk-adjusted basis.

Unified Modeling Language Diagrams



Liquidity-Weighted Trend Strategy



Feature Engineering

Data

The strategy relies primarily on market price data.

Key Variables

- `m_close_dividend_and_split_adjusted` — Adjusted close prices
- Trading volume
- Daily traded value (price × volume)

Adjusted Prices Incorporate

- Dividends
- Stock splits
- Corporate actions

This ensures that signals reflect true total-return dynamics and are not distorted by mechanical price adjustments.

Data Analysis

Time Series and Cross-Sectional Analysis

- US equity prices frequently exhibit trend persistence, supporting the viability of moving-average-based signals.
- Liquidity is concentrated in a small subset of stocks; liquid stocks support scalable systematic strategies.
- The 200-day SMA requires a minimum warmup period, introducing a natural data sufficiency filter.

Key Financial Metrics

- Price trend: direction and duration of SMA crossover states
- Traded value: 3-month rolling average of daily price × volume
- Signal stability: frequency of crossover transitions



Common Visualizations

- Price trend charts with SMA overlays (50-day and 200-day)
- Liquidity ranking distributions across the eligible universe
- Signal distribution over time (fraction of universe in bullish regime)
- Moving average crossover event frequency

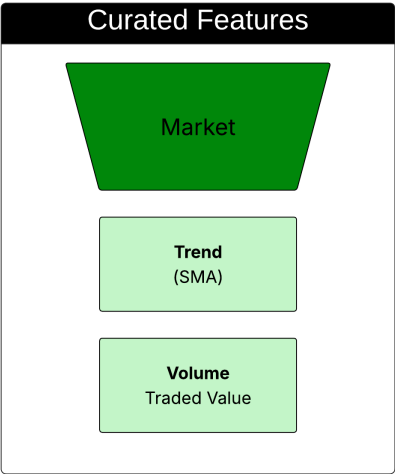
Features

Two core trend features and one liquidity feature are constructed from price data.

Trend Indicators

Feature	Description
c_sma_50d	50-day Simple Moving Average of adjusted close price. First 49 values are NaN (warmup period).
c_sma_200d	200-day Simple Moving Average of adjusted close price. First 199 values are NaN (warmup period).

Note: To avoid overfitting, these features remain fixed throughout the research project.



Features Overview



Strategy Signals

Binary Filter — SMA Crossover Signal

The signal is generated by `c_sma_50d_200d_signal` in `simple_moving_average_alpha_signal.py`:

Signal = 1 when $SMA(50) > SMA(200)$ → Bullish trend (upward momentum)

Signal = 0 when $SMA(50) \leq SMA(200)$ → Bearish trend (downward momentum)

Signal = NaN when insufficient data for either SMA → Warmup period

Signal Logic

- The signal is only activated once both SMAs have valid values (minimum 200 trading days of history per ticker).
- This design prevents false signals during the warmup period.
- The signal captures the classic "Golden Cross" (bullish) and "Death Cross" (bearish) patterns.



Strategy Design

Strategy Modeling

Most research stops at: **"Does the signal predict returns?"**

But strategy design requires answering three deeper questions during construction:

Dimension	Question
Selection	Which stocks are above both their 50-day and 200-day SMAs and rank among the Top 35 by 3-month traded value?
Sizing	How should capital be allocated among selected stocks based on traded value, respecting the 20% maximum per position?
Timing	When does the Top 35 composition change, requiring portfolio rebalancing?

Portfolio Construction

Selection Rules

- Securities with `c_sma_50d_200d_signal = 1` (bullish crossover active)
- Ranked by `c_daily_traded_value_63d` (3-month traded value, descending)
- Top 35 securities selected
- If fewer than 35 eligible securities exist, remaining capital allocated to SPY ETF

Sizing Rules

- Position weights proportional to 3-month traded value
- Maximum weight of 20% per individual position
- Weights normalized to sum to 100% across the portfolio



Rebalancing Logic — Event-Driven

Portfolio changes are triggered when:

- A security enters the Top 35 liquidity ranking with an active signal
- A security exits the Top 35 liquidity ranking or loses its signal
- The number of eligible securities falls below 35 (SPY allocation activated)
- A portfolio component is delisted or becomes non-tradable

Market Regime Handling

During weak markets, fewer stocks may satisfy the trend condition. When eligible stocks < 35, remaining capital is invested in **SPY ETF** to keep the portfolio fully invested, maintain market exposure, and preserve diversification.

Backtesting

Backtesting Config

parameter	value	description
portfolio_name	portfolio_weights	Name of the portfolio
benchmark_file_name	KN_US_Equity_Benchmark	Benchmark name
start_date	2015-01-01	From date, formatted as YYYY-MM-DD; e.g. 1999-12-31
end_date	2026-04-17	To date, formatted as YYYY-MM-DD; e.g. 2000-12-31
initial_capital	\$1,000,000.00	Initial capital to invest
cash_reserve_percentage	1.00%	Percentage of cash to reserve
commission_cents	\$0.05	Cents of every share to charge the transaction costs
input_market_data_directory	Data_Curator	Directory Path where your market data input files are located
input_portfolio_directory	Portfolio_Construction	Directory Path where your portfolio input files are located
backtest_results_output_directory	Backtest_Engine	Directory Path where your backtest result files are located
market_data_input_format	csv	Format of the prices files
portfolio_input_format	csv	Format of the portfolios files
commission_price_column	c_vwap	Column used to calculate the number of shares for commission fees (typically unadjusted VWAP)
trade_execution_price_column	m_vwap_split_adjusted	Column used as the trade execution price for buying/selling shares (typically split/dividend-adjusted VWAP)
mark_to_market_price_column	m_close_dividend_and_split_adjusted	Column used for daily portfolio valuation between rebalancing dates (typically split/dividend-adjusted close)
date_column	m_date	Column containing the trading dates
logger_level	info	Level of detail for the output logs; Shows all logs with severity equal or greater than the selected value
dashboard_port	8050	Port to expose for backtest dashboard



Performance Top 20

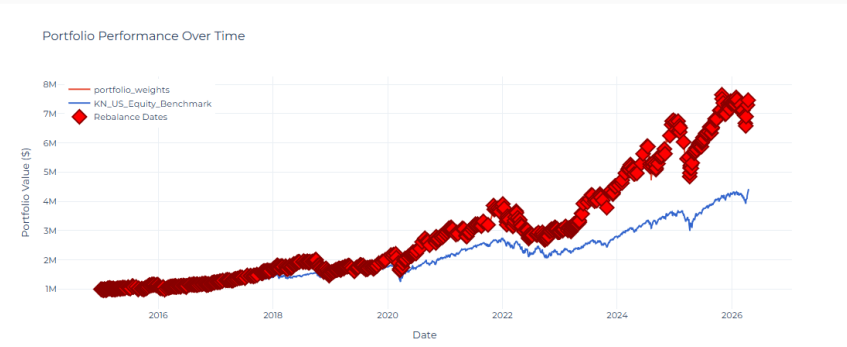
Backtest Dashboard

portfolio_weights

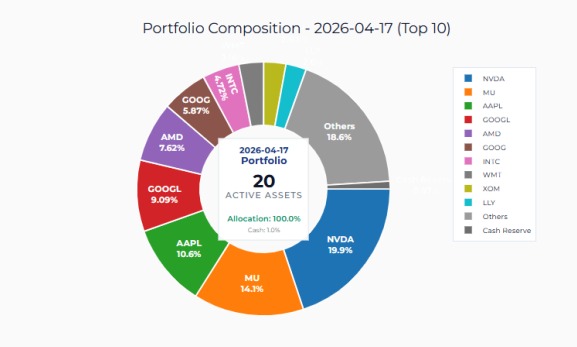
Analysis Period: 2015-01-02 to 2026-04-17



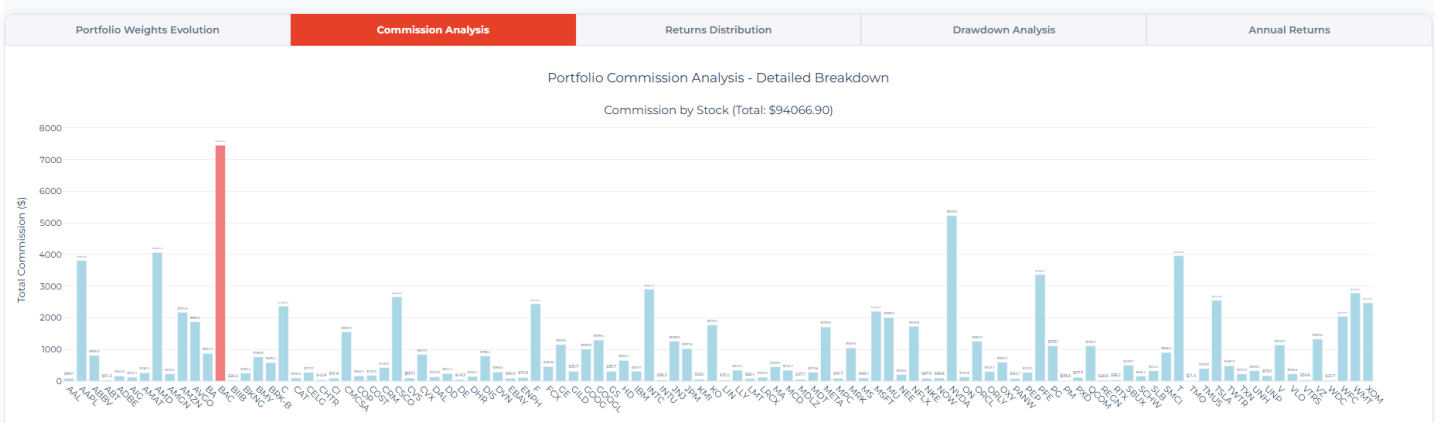
Portfolio Performance



Portfolio Composition

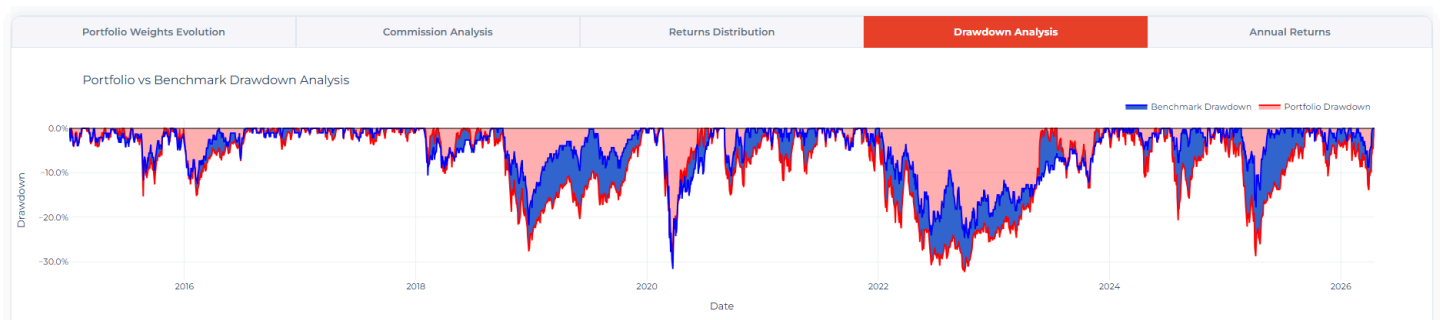


Commissions Top 20





Drawdown Top 20



Annual Returns Top 20

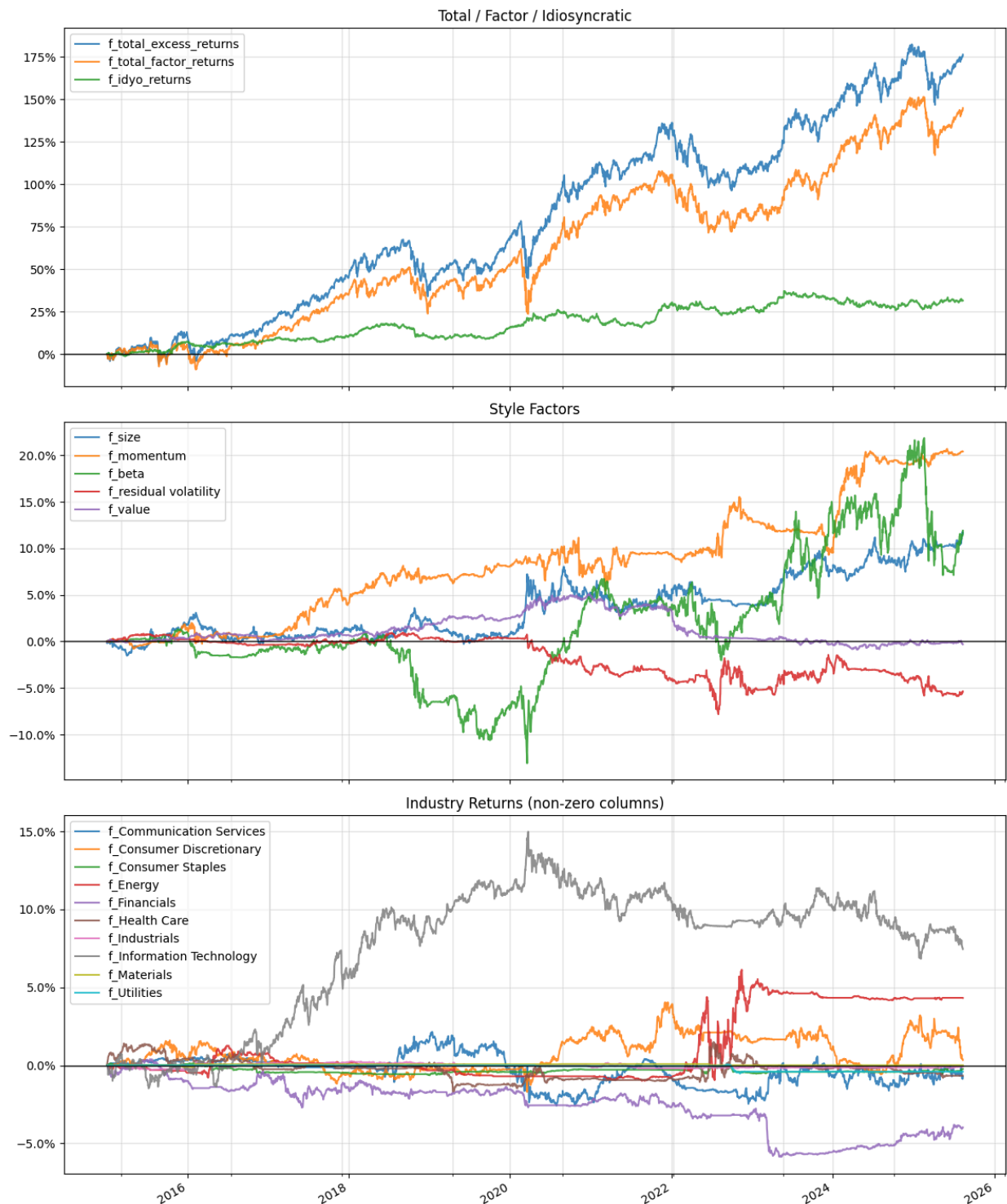
Portfolio Weights Evolution			Commission Analysis		Returns Distribution		Drawdown Analysis		Annual Returns	
Annual Returns Comparison										
Portfolio vs Benchmark Annual Performance										
Year	year	start_date	end_date	portfolio_weights_return	portfolio_weights_start_value	portfolio_weights_end_value	KN_US_Equity_Benchmark_ret	KN_US_Equity_Benchmark_sta	KN_US_Equity_Benchmark_en	alpha
0	2015	2015-01-02	2015-12-31	13.58%	\$999,070.45	\$1,138,786.87	2.35%	\$1,000,000.00	\$1,024,178.34	11.23%
1	2016	2016-01-04	2016-12-30	9.76%	\$1,101,934.00	\$1,212,570.16	12.68%	\$1,008,196.21	\$1,139,771.56	-2.92%
2	2017	2017-01-03	2017-12-29	33.84%	\$1,217,143.91	\$1,640,351.29	21.25%	\$1,149,892.74	\$1,400,709.16	12.58%
3	2018	2018-01-02	2018-12-31	-6.40%	\$1,653,551.51	\$1,544,857.76	-4.70%	\$1,412,429.17	\$1,344,266.15	-1.70%
4	2019	2019-01-02	2019-12-31	31.41%	\$1,537,638.67	\$2,036,056.73	30.01%	\$1,346,253.24	\$1,763,075.43	1.40%
5	2020	2020-01-02	2020-12-31	38.16%	\$2,069,672.46	\$2,888,961.78	17.55%	\$1,779,288.26	\$2,102,226.99	20.62%
6	2021	2021-01-04	2021-12-31	30.96%	\$2,885,232.11	\$3,806,970.41	30.11%	\$2,071,592.52	\$2,715,208.87	0.85%
7	2022	2022-01-03	2022-12-30	-23.62%	\$3,910,959.26	\$2,964,730.79	-19.05%	\$2,733,695.10	\$2,199,926.69	-4.57%
8	2023	2023-01-03	2023-12-29	51.49%	\$2,967,645.22	\$4,540,289.37	27.24%	\$2,192,472.97	\$2,805,823.18	24.24%
9	2024	2024-01-02	2024-12-31	45.19%	\$4,455,659.07	\$6,546,090.45	25.48%	\$2,790,071.13	\$3,526,215.83	19.71%
10	2025	2025-01-02	2025-12-31	13.23%	\$6,450,773.30	\$7,329,642.29	19.63%	\$3,518,945.15	\$4,230,749.15	-6.40%
11	2026	2026-01-02	2026-04-17	15.91%	\$7,367,557.76	\$7,698,567.43	15.29%	\$4,239,874.62	\$4,423,286.18	0.62%

Attribution Analysis

- Factor returns drive nearly all performance.** Of the ~175% cumulative excess return, factor exposures account for ~150%, idiosyncratic (stock-picking) alpha contributes only ~25% and has flatlined since 2022.
- Beta exposure is an unrewarded drag.** The f_{beta} factor hit -12% during the COVID crash and still sits around -5% cumulatively. It adds drawdown risk without compensating return, neutralizing beta (constraining portfolio beta to ~1.0 vs. the benchmark) would immediately improve risk-adjusted performance.
- Size and momentum are the real engines, but momentum needs guardrails.** The size factor steadily contributes ~10-12%, while momentum adds ~5% with sharp reversals in 2020 and 2022.



Adding a momentum-crash filter or regime-aware signal would preserve the upside while reducing those whipsaw drawdowns.





Strategy Score

1. We can improve the coverage of the investable universe using another data provider.
2. We need to be more clear to debug the logic to avoid rebalancing too often when we keep how many days need to pass to change the signal.
3. We can add more attribution analysis results to compare if the strategy actually improves with the ideas we discussed in that section.

Strategy Blueprint	Scoring Concept	Weight	Score (0-100)	Weighted Score	Gap (Score)
Strategy Definition	Hypothesis & Source of Returns	11.11%	100	11.1	0.0
	Research Foundation	11.11%	100	11.1	0.0
	Scientific Process	11.11%	100	11.1	0.0
Feature Engineering	Data Quality & Integrity	11.11%	66	7.3	-3.8
	Feature Design & Relevance	11.11%	100	11.1	0.0
	Signal Construction	11.11%	100	11.1	0.0
Strategy Design	Portfolio Construction Logic	11.11%	66	7.3	-3.8
	Backtesting & Robustness	11.11%	100	11.1	0.0
	Attribution & Understanding	11.11%	66	7.3	-3.8
Overall Score		100.00%		88.7	



Conclusions

Hypothesis Validation.

Did the data support the source of returns?

Key Findings:

- The strategy outperforms the benchmark, confirming the presence of trend persistence
- Performance is primarily driven by factor exposures (momentum + size) rather than pure stock selection
- Returns are not constant across time — periods of underperformance align with known momentum crashes
- Beta exposure introduces unrewarded risk, reducing risk-adjusted performance

Interpretation:

- The hypothesis is partially validated
- The strategy captures real, known drivers of returns
- However, performance is not purely alpha — it is largely explained by systematic exposures

Conclusion:

- The strategy works — but we now understand why it works, and where it breaks

Is This An Implementable Strategy?

Implementation Risks

Factor decay

- Momentum signals weaken as they become crowded

Market impact & liquidity

- Even with liquidity filters, execution costs can erode returns

Turnover & rebalancing

- Event-driven rebalancing may generate unnecessary trading



Capacity constraints

- Strategy scalability is limited to highly liquid securities

Operational Requirements

- Robust data pipelines and execution systems
- Continuous monitoring of signals and exposures
- Strict risk management framework

Decision

- 👉 Suitable for paper trading and controlled deployment

Key Condition

Implementation must include cost control, turnover constraints, and exposure management

Next Steps

1. Strategy Improvements

- Neutralize unrewarded beta exposure
- Introduce momentum crash protection / regime filters
- Enhance signals with additional features (multi-factor approach)

2. Implementation & Infrastructure

- Build production-grade data pipelines
- Implement execution-aware backtesting (costs, slippage)
- Deploy paper trading environment

3. Validation & Monitoring

- Track out-of-sample performance
- Monitor factor exposures and drift
- Define KPIs and alert systems



4. Scaling & Deployment

- Estimate capacity limits and liquidity impact
- Define capital allocation strategy
- Prepare for incremental live deployment



Findings, Concerns and Decisions

Initial Strategy Design

2026-04-28

The SMA crossover (50d/200d) was selected as the primary signal for its simplicity, interpretability, and strong academic backing. The 35-stock liquidity filter was chosen to balance diversification with the capacity limitations of smaller-cap securities. The 20% maximum position cap prevents concentration risk. SPY was selected as the fallback asset to ensure full market exposure during periods of low signal coverage. The warmup period of 200 trading days per ticker must be accounted for when defining the backtest start date to avoid survivorship and look-ahead bias.

2026-04-29

After running the attribution analysis we learned that the strategy drawdowns could be improved trying to neutralize beta and also trying to improve momentum and sizing exposures.